<u>Proof</u> (of Last Theorem) 4/10/23

Suppose S_k, $x \in S_k$ as in the assumptions:

$$c_2 \le \min_{\substack{x \neq 0 \\ x \in S_k}} \frac{x^* A x}{x^* x}$$

$$= \min_{\substack{x \neq 0 \\ x \perp \{u_1, \dots, u_{n-k}\}}} \frac{x^* A x}{x^* x}$$

(where $\{u_1, u_2, \dots, u_{n-k}\}$ is orthonormal basis for $S_k^\perp$ ($\dim S_k = k \Rightarrow \dim S_k^\perp = n-k$)
$((S_k^\perp)^\perp = S_k)$)

$$\le \max_{w_1, \dots, w_{n-k}} \min_{\substack{x \neq 0 \\ x \perp \{w_1, \dots, w_{n-k}\}}} \frac{x^* A x}{x^* x}$$

Courant-Fischer

$$= \lambda_{n-k+1} \le \lambda_{n-k+2} \le \dots \le \lambda_n$$

The other inequality is analogous. ∎

this is what we needed in Sylvester Law

<u>Corollary</u> If $A \in M_n$, $A = A^*$, $x^* A x \geq 0$ $\forall x \in S_k$, S_k = k-dimensional subspace, then A has k nonnegative eigenvalues. If $x^* A x > 0$, then A has k positive eigenvalues

<u>Proof</u> Previous theorem with $C_2 = 0$ ∎

NEXT: CHAPTER 6: LOCATION & PERTURBATION OF EIGENVALUES

An important fact: (Appendix D)

Theorem Let $n \geq 1$ and $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, $a_n \neq 0$, $a_j \in \mathbb{C}$, $j = 1, 2, \ldots, n$. Then, $\forall\, \varepsilon > 0$, $\exists\, \delta > 0$ such that for every polynomial

$$q(x) = b_n x^n + b_{n-1} x^{n-1} + \cdots + b_1 x + b_0$$

$$b_n \neq 0$$

with $\max_{1 \leq i \leq n} |a_i - b_i| < \delta$, we have that

$$\min_{\tau \in S_n} \max_{1 \leq j \leq n} |\lambda_j - \mu_{\tau(j)}| < \varepsilon,$$

permutations

where $\lambda_1, \ldots, \lambda_n$ are the zeros of $p(x)$

$\mu_1, \ldots, \mu_n$ —— " —— $q(x)$

S_n = symmetric group of all permutations.

It says: if you have two polynomials with close enough coefficients, then the zeros (roots) are also close in some pairing.

<u>Proof</u> Classical theorem in algebra of polynomials.

(roots are continuous functions of the coefficients in an algebraic closed field, like $\mathbb{C}$.)

* <u>Corollary</u> Since coefficients of characteristic polynomials are continuous functions of the entries of the matrix (= sums of principal minors), and since (by above theorem) the eigenvalues are continuous function of coefficients, and since composition of continuous functions is continuous, we have

"Eigenvalues are continuous functions of the entries".

§ 6.1 Gersgorin Discs

A good & easy way to locate/localize /estimate location of eigenvalues.

Theorem Let $A = [a_{ij}] \in M_n$. Define

$$R_i'(A) = \sum_{\substack{j=1 \\ j \neq i}}^{n} |a_{ij}|, \quad i = 1, 2, \ldots, n$$

("deleted" row sums of absol. values)

Then

$$\underset{\text{spectrum}}{\sigma(A)} \subseteq \underbrace{\bigcup_{i=1}^{n} \overbrace{\left\{ z \in \mathbb{C} : |z - a_{ii}| \leq R_i'(A) \right\}}^{\text{disc}}}_{= G(A) \text{ (Gersgorin region)}}$$

Furthermore, if the union of k of these n discs in $G(A)$ form a connected region G that is disjoint from the other $n-k$ discs, then G contains exactly k eigenvalues of A.

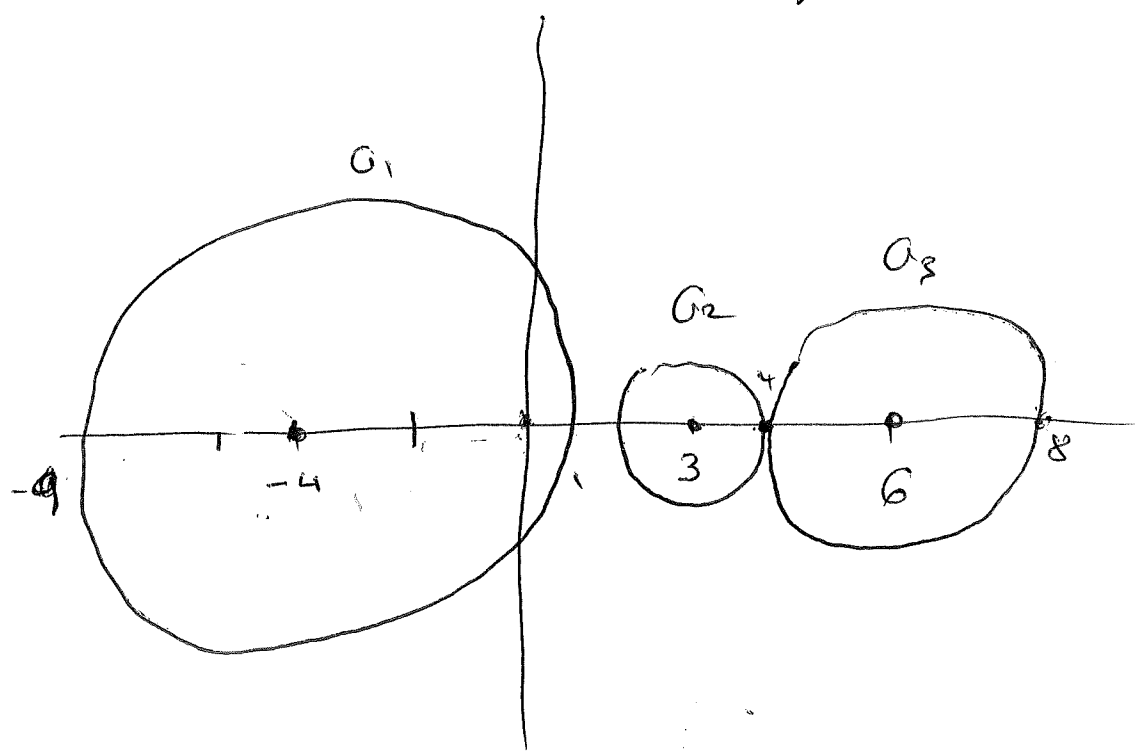
Illustration before proof:

$$A = \begin{bmatrix} 3 & 1 & 0 \\ -1 & -4 & -4 \\ 1 & 1 & 6 \end{bmatrix}$$

$R_1'(A) = |1| = 1$

$R_2'(A) = |-1| + |-4| = 5$

$R_3'(A) = |1| + |1| = 2$



$$G = G_1 \cup G_2 \cup G_3$$

Conclusions based on theorem:

- $\rho(A) \leq 9$
- $G_2 \cup G_3$ is a connected region
  $(G_2 \cup G_3) \cap G_3 = \phi$
  so <u>two</u> eigenvalues in $G_2 \cup G_3$
  and 1 in G_1
- The eigenvalue in G_1 must be real.

4/12/23

<u>Proof</u> (of Gersgorin's Theorem)

Let $Ax = \lambda x$, $x \neq 0$, $x = [x_j]$, $A = [a_{ij}] \in M_n$

Let $x_p \neq 0$ such that $|x_p| \geq |x_j|$, $\forall j$

Then

$$Ax = \lambda x \implies \lambda x_p = [\lambda x]_p = [Ax]_p$$

$$= \sum_{j=1}^{n} a_{pj} x_j$$

$$\implies (\lambda - a_{pp}) x_p = \sum_{\substack{j=1 \\ j \neq p}}^{n} a_{pj} x_j$$

$$\implies |\lambda - a_{pp}| |x_p| \leq \sum_{\substack{j=1 \\ j \neq p}}^{n} |a_{pj}| |x_j|$$

$$\leq |x_p| \underbrace{\sum_{\substack{j=1 \\ j \neq p}}^{n} |a_{pj}|}$$

$$\leq |x_p| R_p'(A)$$

$$\implies |\lambda - a_{pp}| \leq R_p'(A)$$

$$\implies \lambda \in G(A) \implies \sigma(A) \subseteq G(A)$$

(λ lies in the p-th disc)

For the "furthermore" part:

Write $A = D + B$, $D = \mathrm{diag}(a_{11}, a_{22}, \dots, a_{nn})$

Let $A_\varepsilon = D + \varepsilon B$, $\varepsilon \in [0,1]$ $(A = A_1)$

Notice $\underbrace{R_i'(A_\varepsilon)} = R_i'(\varepsilon B) = \underbrace{\varepsilon R_i'(A)}$

otherwise PAP^T ← WLOG suppose

$$G_k := \bigcup_{i=1}^{k} \{ z \in \mathbb{C} : |z - a_{ii}| \le R_i'(A) \}$$

is a connected region that is disjoint from

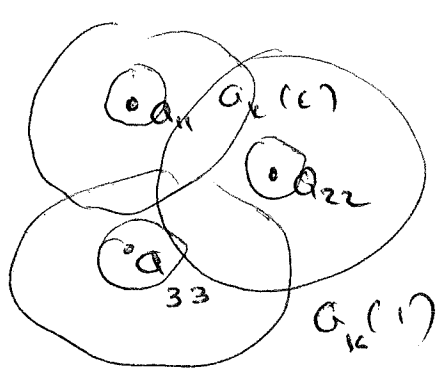
$$G_k^c = G(A) \setminus G_k \qquad (G_k \cap G_k^c = \phi)$$

Note that $\forall\, \varepsilon \in [0,1]$, then

$$G_k(\varepsilon) = \bigcup_{i=1}^{k} \{ z \in \mathbb{C} : |z - a_{ii}| \le R_i'(A_\varepsilon) = \varepsilon R_i'(A) \}$$

$$\subseteq G_k = G_k(1)$$

$k=3$



For each $i = 1, 2, \ldots, k$, consider $\lambda_i(A_0) = \lambda_i(D)$
$= a_{ii}$

and also consider $\lambda_i(A_\varepsilon)$ as a function of $\varepsilon \in [0,1]$

By continuity of eigenvalues as functions of entries and because

$$\lambda_i(A_\varepsilon) \in G_k(\varepsilon) \subseteq G_k, \ \forall \varepsilon \in [0,1]$$

we have that $\lambda_i(A_0) = a_{ii}$ is joined to $\lambda_i(A_1) = \lambda_i \in \sigma(A)$ by a continuous curve that lives in G_k

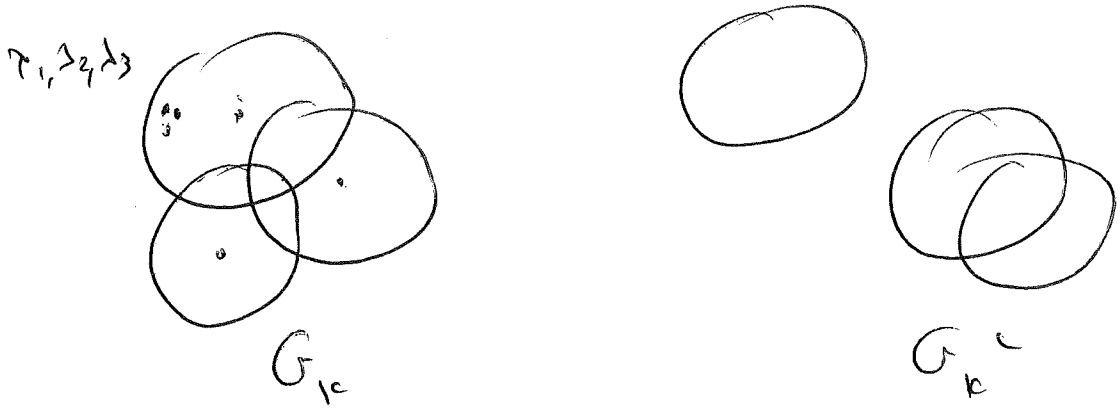
So $\forall \varepsilon \in [0,1]$, there at least k eigenvalues of A_ε in $G_k(\varepsilon)$

But this means there must be <u>exactly</u> k eigenvalues of A_ε in $G_k(\varepsilon)$ because the same continuity argument shows at least $n-k$ eigenvalues live in $G_k^c(\varepsilon)$.

(cannot jump because $G_k^{(c)} \cap G(\varepsilon) = \emptyset$.

Thus G_k contains k eigenvalues of A. ∎

<u>Note</u> it can be that



must contain 3 eigenvalues

no word as to where they are.

They can be all in one of the 3 discs.

<u>Find an example!</u>

<u>Corollary</u> Since $\sigma(A) = \sigma(A^T)$, we have

$$\sigma(A) \subseteq \underbrace{G(A) \cap G(A^T)}_{\text{gives better estimates!}}$$

<u>Definition</u> We call $A=[a_{ij}] \in M_n$ <u>diagonally dominant</u> if

$$|a_{ii}| \geq R_i(A) = \sum_{\substack{j=1 \\ j \neq i}}^{n} |a_{ij}|$$

for each $i=1,2,\dots,n$.

(<u>strictly</u> <u>diag.</u> <u>dominant</u> if all $\geq \longrightarrow >$)

<u>Theorem</u> Let A be strictly diagonally dominant. Then

(i) A is invertible.

(ii) if $a_{ii} > 0$ for all i, then

$$\operatorname{Re} \sigma(A) \subseteq \mathbb{R}_+ \quad \text{(positive reals)}$$

(iii) if $A = A^*$ & $a_{ii} > 0$, then $\sigma(A) \subseteq \mathbb{R}_+$ (i.e. A is positive definite).

<u>Proof</u> Apply Gersgorin to A.

$$G(A) \cap \{0\} = \emptyset$$

(known before Gersgorin as Levy-Desplanque)

4/14/23

§ 6.3 Perturbation Theorems (eigenvalues)

Motivation $D = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n) \in M_n$

Perturb D by $E = [e_{ij}] \in M_n$. What can we say about $\sigma(D+E)$?

By Gersgorin

$$\sigma(D+E) \subseteq \bigcup_{i=1}^{n} \left\{ z \in \mathbb{C} : |z - \underline{\lambda_i - e_{ii}}| \le \sum_{\substack{j=1 \\ j \ne i}}^{n} |e_{ij}| \right\}$$

$$\subseteq \bigcup_{i=1}^{n} \left\{ z \in \mathbb{C} : |z - \lambda_i| \le \sum_{j=1}^{n} |e_{ij}| \right\}$$

(used $\big||a-b| - |c|\big| \le |a-b-c|$)

($\big||x| - |y|\big| \le |x-y|$)

So if $\hat{\lambda} \in \sigma(D+E)$, then there exists $\lambda_i \in \sigma(D)$ such that

$$|\hat{\lambda} - \lambda_i| \le \sum_{j=1}^{n} |e_{ij}|$$

$$\le \max_i \sum_{j=1}^{n} |e_{ij}| \overset{!}{=} |||E|||_\infty$$

Wonderful: Eigenvalues of the perturbation we'll study. $D+E$ can be close to $\sigma(D)$ if E is "small".

For general matrices, situation is less clear.

We will need a few basics on matrix norms:

<u>Definition</u> Function $|||\cdot||| : M_n \longrightarrow \mathbb{R}$ is a <u>matrix norm</u> provided that

(1) $|||A||| \geq 0$

(2) $|||A||| = 0 \iff A = 0$

(3) $|||cA||| = |c|\,|||A|||, \ \forall c \in \mathbb{C}$ (homogeneous)

(4) $|||A+B||| \leq |||A||| + |||B|||$ (triangle)

optional $\longleftarrow$ (5) $|||AB||| \leq |||A|||\,|||B|||$

for all $A, B \in M_n$.

If (5) is satisfied, we call $|||\cdot|||$ a <u>submultiplicative</u>.

<u>Examples</u> (commonly used matrix norms) $A = [a_{ij}] \in M_n$

$$\|A\|_1 = \sum_{i,j=1}^{n} |a_{ij}| \quad (\ell_1\text{-norm})$$

$$\|A\|_2 = \left(\sum_{i,j=1}^{n} |a_{ij}|^2\right)^{1/2} = \|A\|_F$$

(ℓ_2-norm, Euclidean matrix norm, Frobenius norm)

$$\|A\|_\infty = \max_{i,j} |a_{ij}| \quad (\ell_\infty\text{-norm})$$

(we used $\|\cdot\|$ for the above norms to distinguish them from matrix norms induced by vector norms as follows.)

<u>Definition</u> If $\|\cdot\| : \mathbb{C}^n \to \mathbb{R}$ we denote a vector norm, then the matrix norm defined by

$$|||A||| := \max_{\substack{x \in \mathbb{C}^n \\ x \neq 0}} \frac{\|Ax\|}{\|x\|} = \max_{\|x\|=1} \|Ax\|$$

is called a matrix norm <u>induced</u> by the given vector norm.

(Can be shown that "induced norm" are indeed matrix norms)

<u>Facts</u> For induced norms:

- $|||I||| = 1$
- $\|Ax\| \leq |||A||| \cdot \|x\|$

  (easy to prove)

<u>Examples</u> (of induced matrix norms)

(1) $$|||A|||_1 = \max_j \sum_{i=1}^{n} |a_{ij}| \quad \text{(max column sum)}$$

is induced by the vector ℓ_1-norm:

$$\|x\|_1 = |x_1| + |x_2| + \dots + |x_n|$$

(2) $$|||A|||_\infty = \max_i \sum_{j=1}^{n} |a_{ij}| \quad \text{(max row sum)}$$

is induced by ℓ_∞-norm:

$$\|x\|_\infty = \max\{|x_1|, |x_2|, \dots, |x_n|\}$$

(3) $$|||A|||_2 = \max\{\sqrt{\lambda} : \lambda \in \sigma(A^*A)\}$$

is induced by Euclidean norm

$$\|x\|_2 = \left(\sum_{j=1}^{n} |x_j|^2\right)^{1/2}$$

(proof: Singular Value Decomposition)

($|||\cdot|||_2$ is known as the <u>spectral</u> norm)

<u>Theorem</u> Let $|||\cdot|||$ be any submultiplicative matrix norm. Then for every $A \in M_n$,

$$\rho(A) = \max\{|\lambda| : \lambda \in \sigma(A)\} \leq |||A|||.$$

$$\left(\text{related: } \rho(A) = \lim_{k\to\infty} |||A^k|||^{1/k}\right)$$